

Apache Spark

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Outline

What is Apache Spark?

Practical Features of Apache Spark

Your turn to try!

Overview of Apache Spark

- ▶ Apache Spark is a multi-language engine for executing data engineering, data science, and machine learning on single machines or clusters.
- ▶ It is a powerful tool which is made for large-scale amounts of information.
- ▶ As we have talked about in the prior presentation on Big Data Analytics, Apache Spark is fast due to its use of in-memory processing, its resilient distributed datasets (which are fault tolerant!), and DAGs for execution.

Querying data with Apache Spark

- ▶ Spark has a large list of built-in functions which allow you to query data.
- ▶ All queries begin with the dataframe you are accessing, followed by the functions you wish to call.
- ▶ Functions include items like `.groupBy`, `.orderBy`, `.limit`, `.sum` to name a few.

Access the practical section

- ▶ Access the PySparkExamples.ipynb on my github and open it in collab.
- ▶ <https://github.com/b-pinter/miniLectures>
- ▶ <https://spark.apache.org/docs/latest/api/sql/index.html>